

Pratik Shrivastava

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WORK EXPERIENCE

- Dodge AI**
Machine Learning Intern

Jun. 2025 – Jul. 2025
Remote, KA

 - Built a purchase-intent classifier on 2.75M+ events (22 leakage-safe RFM features), reaching 0.156 PR-AUC ~2x baseline – on a 0.2%-positive class.
 - Shipped a hybrid recommender (content + ALS + cold-start) scoring ~24x the MAP@10 of a most-popular baseline on held-out users.
 - Used 16-factor ALS embeddings instead of a dense item-similarity matrix (infeasible at 185K products), scaling coverage from 5K to the full catalog.
- Siemens Technology & Services Pvt Ltd.**
Software Engineering Intern

May. 2024 – Aug. 2024
Remote, KA

 - Built RESTful API integration layer for a 3-tier IIoT water-monitoring dashboard with near real-time polling and incremental UI updates.
 - Designed GIS map visualizations with colour-coded, per-sensor status markers for quick anomaly spotting.
 - Handled missing/inconsistent sensor data gracefully in the parsing layer to keep visualizations from breaking.

PROJECTS

- Drape**
Python, FastAPI, DINOv2, FAISS, ViT

[Project Link](#)

 - Built an image-to-image visual fashion search engine over a 20K-item catalog using YOLOv8 detection + DINOv2 embeddings (768-d) + FAISS search, achieving sub-10ms retrieval latency.
 - Designed a neighbor-consensus system generating product descriptions and retailer links directly from retrieval results cutting substantial API calls per search to a handful of parallel fetches.
 - Identified a data-integrity flaw in a public benchmark dataset via cross-tabulation analysis, excluding non-unique identifiers from the 20K-item catalog before they could corrupt product metadata.
- Harmoniq**
Python, FastAPI, Scikit-learn, KNN, Numpy, Pandas

[Project Link](#)

 - Built a content-based music recommender over an 89K track Spotify catalog: cosine KNN on a 9-D audio feature embedding.
 - Engineered the data pipeline – deduplicated ~114K rows to 89K tracks and dropped an autoencoder after measuring that a standardized-feature space gave better cosine-ranking resolution.
 - Designed an explainable ranking layer that scores every recommendation on 6 weighted signals and surfaces the closest audio feature as a plain-English reason, with per-artist diversity caps.
- Harmoniq**
Python, FastAPI, Scikit-learn, KNN, Numpy, Pandas

[Project Link](#)

 - Built a 5-stage PDF-to-illustrated-book pipeline (multi-extractor text parsing + OCR, LLM character extraction, prompt generation, image rendering) driven by a Postgres-backend job queue with pause/resume and restart recovery.
 - Engineered a “visual bible” for cross-page character consistency: union-find alias grouping + single-pass trait extraction inlined into every image prompt.
 - Abstracted all AI behind 3 pluggable provider layers (LLM/images/storage, 9 backends), so identical code runs on a local GPU using Diffusion models or free tier deployment.

EDUCATION

- Manipal University**
Bachelor of Technology in Computer Science & Engineering

Aug. 2022 – Aug 2026
Jaipur, Rajasthan

SKILLS

Programming Languages	Python, C++, JavaScript, SQL
Frameworks & Libraries	Node.js, RESTful APIs, FastAPI, Express.js
Databases	MySQL, PostgreSQL, Supabase BaaS
ML	Transformers, ViTs, VLMs, Pytorch, Pandas, Tensorflow, Keras, Matplotlib, Numpy, Pandas
Developer Tools	Git, Github, Postman, VS Code